

Explainable AI for Healthcare Predictive Models

1. Introduction

This report summarises the process and findings of applying machine learning models and Explainable AI (XAI) techniques to the UCI Heart Disease Dataset (Cleveland). The primary goal was to build predictive models for heart disease presence and understand their decisions using SHAP and LIME.

2. Dataset and Preprocessing

- **Dataset:** UCI Heart Disease Dataset - Cleveland.
- **Task:** Binary classification (0 = No Disease, >0 = Disease).
- **Preprocessing Steps:**
 - Handling missing values (using '?' as na_values).
 - Duplicate rows removed.
 - Data converted to numeric types.
 - Rigorous imputation: Median for continuous features (age, trestbps, chol, thalach, oldpeak) and mode for categorical features (sex, cp, fbs, restecg, exang, slope, ca, thal).
 - Target variable binarized to 0 (absent) or 1 (present).
 - The cleaned dataset contains 303 rows and 14 columns, with no remaining missing values.

3. Exploratory Data Analysis (EDA)

- The target variable showed a relatively balanced distribution, with slightly more patients having heart disease.
- A correlation matrix was generated to visualize relationships between clinical variables.

4. Model Training and Evaluation

Four machine learning models were trained and evaluated on a stratified 80/20 train-test split (SEED=42). Standard scaling was applied within pipelines for linear/kernel models to prevent data leakage.

Model Performance Summary:

Model	Accuracy	Precision	Recall	F1	ROC_AUC
Random Forest	0.8852	0.8387	0.9286	0.8814	0.9545
XGBoost	0.8852	0.8387	0.9286	0.8814	0.9535
Logistic Regression	0.8689	0.8125	0.9286	0.8667	0.9513
SVM	0.8525	0.8065	0.8929	0.8475	0.9437

- **Best Model:** The **Random Forest** model achieved the highest ROC-AUC score of **0.9545**, indicating excellent discriminative power.
- Performance comparison plots and ROC curves were generated to visually represent these results.

5. Explainable AI (XAI)

SHAP and LIME were used to explain the predictions of the best-performing model, Random Forest.

5.1 SHAP (SHapley Additive exPlanations)

- **Global Explanation:** A SHAP summary plot identified the most important features influencing the model's overall predictions.
- **Local Explanation:** A waterfall plot provided a patient-level explanation, detailing how each feature contributed to the prediction for a specific patient (e.g., patient at sample_index = 0).
- **Top Features (Global Importance):** thal, ca, and cp were consistently identified as highly influential.

5.2 LIME (Local Interpretable Model-agnostic Explanations)

- **Local Explanation:** A LIME explanation for the same patient provided a human-interpretable local explanation, highlighting features important for that specific prediction.
- **Aggregated Importance:** LIME feature importance was aggregated across multiple samples, showing overall feature relevance.

5.3 SHAP vs LIME Agreement and Stability

- **Spearman Rank Correlation:** The agreement between SHAP and LIME feature rankings was very high, with a Spearman rank correlation of **0.9780**.
- **Top-10 Feature Agreement:** All top 10 features identified by SHAP were also among the top 10 identified by LIME, resulting in a **1.0000** agreement score.
- **SHAP Stability:** The overall SHAP explanation stability score, assessed by repeating feature importance calculations on bootstrapped samples, was **1.0000**, indicating high consistency in explanations.

6. Conclusion

The project successfully developed and evaluated machine learning models for heart disease prediction. The Random Forest classifier emerged as the best model. Critically, both SHAP and LIME provided consistent and stable explanations, reinforcing confidence in the model's interpretability. The high agreement between these two distinct XAI methods strengthens the validity of the identified key features (e.g., thal, ca, cp) in predicting heart disease presence.

All generated plots, models, and results were saved to their respective directories for further review and deployment.

Report of the Entire Code

Overview

This notebook performs an Explainable AI (XAI) analysis for healthcare predictive models using the UCI Heart Disease Dataset (Cleveland). It covers the full machine learning pipeline from data preparation to model training, evaluation, and interpretation using SHAP and LIME. The goal is to build accurate models for heart disease prediction and provide insights into their decision-making processes, as well as compare the consistency of different XAI methods.

Key Steps and Components

1. Environment Setup and Reproducibility:

- Installs necessary packages (shap, lime, xgboost, joblib).
- Imports essential libraries for data manipulation (pandas, numpy), visualization (matplotlib, seaborn), machine learning (sklearn, xgboost), and XAI (shap, lime).
- Sets a random seed for reproducibility.
- Creates project directories (data, plots, models, results, explanations) to organize outputs.

2. Data Loading and Preprocessing:

- **Environment-Agnostic Loading:** Handles data loading for both Google Colab and local environments, expecting a file named `processed.cleveland.data`.
- **Column Definition:** Defines 14 column names for the dataset, including `age`, `sex`, `cp`, `trestbps`, `chol`, `fbs`, `restecg`, `thalach`, `exang`, `oldpeak`, `slope`, `ca`, `thal`, and `target`.
- **Missing Value Handling:** Loads the dataset, treating '?' as missing values. Removes duplicate rows.
- **Data Type Conversion:** Converts all columns to numeric, coercing errors.
- **Imputation:** Rigorously imputes missing values:
 - Median imputation for continuous variables (`age`, `trestbps`, `chol`, `thalach`, `oldpeak`).
 - Mode imputation for categorical/ordinal variables (`sex`, `cp`, `fbs`, `restecg`, `exang`, `slope`, `ca`, `thal`).
 - Drops rows where the 'target' variable is missing.
- **Target Binarization:** Transforms the target variable from 0-4 to a binary (0 or 1) classification problem, where `>0` indicates the presence of heart disease.
- **Saving Clean Data:** Saves the processed dataset as `heart_disease_cleveland_clean.csv`.

3. Exploratory Data Analysis (EDA):

- Generates descriptive statistics and saves them to `results/descriptive_statistics.csv`.
- **Class Distribution:** Visualizes the distribution of the target variable using a countplot, saved as `plots/class_distribution.png`.
- **Correlation Matrix:** Displays a heatmap of the correlation matrix between all clinical variables, saved as `plots/correlation_matrix.png`.

4. Model Training and Evaluation:

- **Train/Test Split:** Splits the data into 80% training and 20% testing sets, stratified to maintain class proportions (`random_state=42`).
- **Model Definitions:** Defines four classification models:

- Logistic Regression (with StandardScaler pipeline)
- Random Forest Classifier
- Support Vector Machine (SVM) (with StandardScaler pipeline)
- XGBoost Classifier
- **Model Training:** Trains each model on the training data and saves the trained models as .joblib files in the models directory.
- **Model Evaluation:** Calculates and stores performance metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC) for each model on the test set. The results are saved as results/model_performance.csv.
- **Performance Visualization:** Generates bar charts comparing model performance metrics and ROC curves, saved as plots/model_performance_comparison.png and plots/roc_curves.png.

5. Explainable AI (XAI):

- **SHAP (SHapley Additive exPlanations) - Random Forest:**
 - **Global Explanation:** Calculates SHAP values for the Random Forest model and generates a SHAP summary plot to show global feature importance, saved as plots/shap_random_forest_summary.png. It also saves the mean absolute SHAP values to results/shap_random_forest_feature_importance.csv.
 - **Local Explanation:** Provides a waterfall plot for a specific patient's prediction, illustrating how each feature contributes to that individual prediction, saved as plots/shap_local_patient.png.
- **LIME (Local Interpretable Model-agnostic Explanations):**
 - **Local Explanation:** Creates a LIME explainer and generates a local explanation for the same specific patient as SHAP, saved as plots/lime_random_forest_patient.png.
 - **Aggregated LIME Importance:** Aggregates LIME feature importance across multiple test samples to derive a more global understanding of feature relevance.

6. XAI Agreement and Stability Analysis:

- **SHAP vs. LIME Agreement:** Compares SHAP and LIME feature rankings and calculates the Spearman rank correlation to quantify the agreement between the two methods. Also calculates the agreement percentage of top-N features.
- **SHAP Explanation Stability:** Assesses the stability of SHAP explanations by repeating feature importance calculations on bootstrapped samples and computes an overall Spearman stability score.

7. Final Output and Summary:

- Identifies the best-performing model based on ROC-AUC.
- Generates a project summary file (results/project_summary.txt) detailing the best model, its AUC, SHAP stability, SHAP-LIME rank correlation, and common top features.
- Prints a final completion message with key results.

Findings

- The Random Forest model achieved the highest ROC-AUC of 0.9545, indicating strong predictive performance.
- Both SHAP and LIME consistently identified similar key features. The Spearman rank correlation between their feature rankings was very high (0.9780), and the top-10 features showed 100% agreement.
- SHAP explanations demonstrated high stability with a score of 1.0000, suggesting robust and consistent interpretations.
- Features like thal, ca, and cp were consistently highlighted as highly influential across both models and XAI techniques.

Conclusion

The project successfully demonstrates a comprehensive approach to building and interpreting machine learning models for heart disease prediction. The high agreement and stability observed between SHAP and LIME reinforce the trustworthiness of the model's explanations, providing valuable insights into the factors contributing to heart disease predictions.
